Department of Information Systems
Bachelor of Information Systems
Lab Activity 4

IS312 Web Application Development

Weighting:	3%
ULO addressed:	ii, iii, vi, vii
Due date:	11:59 PM, Monday 28th April 2025
Total Marks:	30

Web Application Architecture

System Design Challenge – DWU Student Portal (Moodle)

This task enhances comprehension of the structural layout of modern web applications and how different tiers communicate.

Objective: Design a basic web application architecture using tiers.

Activity:

In you group sketch a diagram of DWU Learning Management System (LMS) application (commonly known as Moodle) in 3-tier or N-tier format.

Identify and label each tier: Front-end (Presentation), Application (Logic), Data (Resources). Present and explain the flow of data between layers.

Identify and list the components exist in each layer (PHP Scripts in application, MySQL in data, HTML/JS in front-end).

Presentation:

Prepare a 5-minute presentation explaining:

- Flow of data
- Technology choices
- Security or performance considerations
- Benefits of the chosen architecture

Tools:

- PowerPoint for presentation
- Lucid Chart, Draw.io, Signavio or any other tool to sketch application 3-tiers or N-tier application portal for the Moodle LMS System

Marking Criteria

Criteria	Excellent (5)	Good (4)	Fair (3)	Poor (1–2)	Mark
Architecture Diagram	Clearly structured, well-labelled tiers and interactions; accurate representation	Mostly clear structure; minor labeling or tier placement issues	Partially clear; missing labels or confused tiers	Unclear or incomplete diagram	5
Explanation of Data Flow	Fully explains how data moves across all tiers; uses examples	Explains basic data flow; may miss one tier or an example	Some attempt at explaining flow; confusing or incomplete	Minimal or incorrect explanation	5
Technology Justification	Excellent rationale for tool/tech use at each layer	Justifies most tools with reasonable arguments	Limited tech explanation	No mention or wrong tool references	5
Team Collaboration	Evident teamwork, all members participate	Good effort, most members contribute	One/two students dominate	No team balance, poor collaboration	5
Presentation Quality	Clear, engaging, well-paced, answers all questions	Generally clear, minor issues in delivery	Hard to follow at times	Disorganized or mumbled	5
Creativity & Depth	Adds additional layers, real-world examples, or unique enhancements	Some creative elements	Basic application only	Minimal effort shown	5
Total					30